Web Application Framework (WAF) - Fall 2026
Task 3 (Deadline: 22nd October 2026)

Develop a web application using Node.js, Express.js, and EJS (Embedded JavaScript) that displays data in an HTML page and allows users to perform four operations: Add, Remove, Edit, and Update. The application should first fetch data from a file containing an array of objects and display records in an HTML page. Four buttons will be provided to manage each record.

View All: Retrieves all records and displays them.

Add: Adds a new record to the view HTML dynamic page interface. When this button is clicked, a page containing form data should be displayed and injected into the array of objects and later rendered on the interface of the HTML page.

Remove: Deletes a selected record from the interface.

Edit: Allows the user to select a record and modify its information.

Update: Saves the edited information and displays the updated record in the view.

The application should use Express.js to handle routing and server-side logic, while EJS renders the dynamic HTML pages. The purpose is to understand CRUD operations and how a template engine works with an Express.js application. In addition, add the following middleware:

Request Logging Middleware: logs HTTP method, URL, timestamp, request duration

- Log each incoming request to a file
- Record the method, URL, date/time, and processing time.
- Log CRUD actions such as adding, editing, and removing records.

Authorization Middleware: user roles, such as admin and viewer, to restrict operations

- **Admin:** Can view, add, edit, update, and remove records.
- **Viewer:** Can only view records.

Input Validation Middleware:

- Validates the fields before adding or updating records

Rate Limiting Middleware:

- Limit requests from a single IP address within a specified time window.
- Allow trusted IPs while blocking known abusive IP addresses.
- Limit repeated requests to delete operations.
- Display an error page when a limit is exceeded.

Error Handling Middleware: missing records, invalid record IDs, handles invalid routes, file errors, validation failures, and unexpected server errors.